
Dark Sparse-Autoencoder Features in Protein Language Models Encode Sub-Domain Motifs, Not Unknown Biology: A Quantitative Audit

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Abstract

Sparse autoencoders (SAEs) trained on protein language model (PLM) activations decompose each layer into thousands of latent features, of which roughly 80% do not align with any SWISS-PROT concept. These *dark features* have been hypothesized to encode biological mechanisms not yet catalogued by experimentalists, but no prior work tests this claim against causal, functional, or literature-grounded criteria at once. We audit the top dark features of the pretrained INTERPLM SAE for ESM-2-8M (layer 4, 10,240 latents) on 1,500 human SWISS-PROT proteins (422K residues) along three axes: (i) statistical structure against a random-residue null, (ii) causal influence on ESM-2-8M’s masked-LM logits via forward-hook ablation, and (iii) post-hoc annotation by GPT-5 prompted on 15 maximally-activating sequence windows per feature. Eight of the top ten dark features detect the TGEKP inter-finger linker of C2H2 zinc fingers, and the remaining two detect the DRY and NPxxY motifs of class-A GPCRs; adding “Zinc finger” to the concept list lifts seven of those eight to $F1 \geq 0.6$. Single-feature ablation, however, fails for *all* 30 top dark features under Benjamini–Hochberg FDR control: KL divergence at activating residues is on average $2^{-3.86} \times$ *smaller* than at matched control residues. Our results refine the prevailing “dark feature = unknown biology” framing: in ESM-2-8M, top dark features encode well-known sub-domain motifs that SWISS-PROT describes at coarser granularity, and high feature–concept correlation does not imply causal mechanistic role under single-feature ablation. We release the pipeline (SAE extraction $\rightarrow$ dark filter $\rightarrow$ PLM-grounded annotation) as a reusable audit tool for future PLM interpretability work.

1 Introduction

Protein language models such as ESM-2-8M [Lin et al., 2023] predict structure and variant fitness without ever seeing a structural label, which strongly suggests their hidden states encode biology. Mechanistic-interpretability work on these models has converged on a recipe borrowed from large language models: train a sparse autoencoder (SAE) on per-residue activations and inventory the resulting features. The flagship INTERPLM study [Simon and Zou, 2024] reports roughly 2,500 SAE features per layer with $F1 > 0.5$ against at least one SWISS-PROT concept, against only 46 such units in the dense neuron basis. The headline observation, however, is that *only* $\sim 20\%$ of SAE features map cleanly to any annotated concept. The remaining *dark features* have repeatedly been described as candidate “model-generated scientific hypotheses”: features that fire consistently on protein residues but that no biologist has yet named.

Why this matters. If even a small fraction of dark features encode real biology, they form a new pipeline for discovery, parallel to BLAST-style sequence search and to AlphaFold-driven structural inference. If most of them are noise, the field’s enthusiasm for SAE-based “interpretability” on

PLMs is overstated. Either answer is actionable, and to our knowledge the question has not been answered with quantitative tests that go beyond feature–concept correlation.

Gap. Existing audits of SAE features in PLMs rely almost entirely on feature–concept F1. Three weaknesses follow. (i) F1 is a *correlational* test: a feature with high concept F1 may not causally influence the model’s predictions, as recent work on antibody PLMs has shown for top- k SAE features [Haque et al., 2026] and as the broader “interpretability without actionability” critique argues for LLMs [Bricken et al., 2026]. (ii) F1 against a finite concept list cannot tell apart noise from concepts the list does not include—in SWISS-PROT’s case, sub-domain motifs (e.g., the TGEKP linker between consecutive C2H2 zinc fingers, or the DRY and NPxxY motifs internal to class-A GPCRs) are absorbed into much coarser parent features. (iii) Nobody has paired SAE feature inventories with experimental fitness measurements at the per-residue level to test whether dark features track functional sensitivity. Our work fills these three gaps in one study.

Our approach. We instrument the pretrained INTERPLM SAE for ESM-2-8M layer 4 on a 1,500-protein random subset of the human SWISS-PROT proteome (422,146 residues) and run a three-axis audit of its dark features. We test (i) whether dark features have statistically more structured activation patterns than a random-residue null; (ii) whether they are causally important to ESM-2-8M’s own masked-LM logits via forward-hook ablation paired with a non-activating control; and (iii) whether GPT-5, prompted with 15 sequence windows per feature, can recover a coherent biological hypothesis. We close the loop by extending the concept list and re-checking F1 against the LLM-proposed labels.

Findings, in numbers. The top 10 dark features under our pipeline produce 10/10 specific, named hypotheses from GPT-5 (9 high-confidence). Eight identify the TGEKP inter-finger linker of C2H2 zinc fingers; two identify the DRY and NPxxY motifs of class-A GPCRs. Adding “Zinc finger” to the F1 evaluation lifts 7 of the 8 features above $F1 = 0.6$ (max 0.84), confirming the LLM hypotheses against the same evaluation that originally tagged them as dark. *Causal ablation fails for all 30 top dark features:* the median $\log_2$ -ratio of $KL(\text{orig}||\text{ablated})$ at activating vs. matched control residues is -3.86 , and zero features survive Benjamini–Hochberg $FDR < 0.10$ [Benjamini and Hochberg, 1995]. Across the broader population of 1,207 “structured dark” features, even after adding 7 additional concepts, $\sim 97\%$ remain dark at $F1 \geq 0.5$, consistent with a long tail of sub-domain motifs that no current annotation system encodes.

Contributions.

- We provide the first quantitative, three-axis audit of “dark” SAE features in a PLM, combining a structured-vs-random null test, causal forward-hook ablation, and LLM-grounded annotation on the same feature set.
- We show that the top dark features in ESM-2-8M encode *sub-domain motifs that SWISS-PROT describes at coarser granularity* (TGEKP, DRY, NPxxY), not unknown biology, and that adding a single concept (“Zinc finger”) re-claims 7 of 8 features at $F1 \geq 0.6$.
- We deliver a quantitative negative result for the “feature–concept F1 implies mechanism” assumption: none of 30 top dark features pass single-feature ablation under FDR control, and the median KL effect is in fact $2^{-3.86} \times$ *smaller* at activating residues than at matched controls.
- We release the audit pipeline (SAE extraction $\rightarrow$ dark filter $\rightarrow$ entropy null $\rightarrow$ ablation $\rightarrow$ GPT-5 annotation $\rightarrow$ concept re-check), runnable end-to-end on a single A6000 in ~ 25 minutes for any INTERPLM checkpoint, as a reusable tool for future PLM interpretability work.

Organization. section 2 positions our work against PLM interpretability and the “actionability” critique. section 3 describes data, model, SAE, and each audit axis. section 4 reports the five experiments. section 5 draws implications for PLM-driven discovery. section 6 concludes with concrete next steps.

2 Related Work

Sparse autoencoders for protein language models. Interpretability of PLMs began with attention-based studies showing that specific heads attend to contact maps and binding sites [Vig et al., 2020, Stark et al., 2023]. The current wave applies SAEs borrowed from LLM mechanistic interpretability. INTERPLM [Simon and Zou, 2024] trains a 10,240-latent ReLU SAE per ESM-2-8M layer and reports that mid-layer features cover 143 of 433 SWISS-PROT concepts at $F1 > 0.5$. Inde-

pendent replication on a different model family appears in Adams et al. [2025]. Villegas Garcia and Ansuini [2025] refine layer selection by intrinsic-dimension plateau and demonstrate that zinc-finger features can be steered. Tsui et al. [2025] re-position SAEs from interpretation to prediction: a top- k SAE trained on a LoRA-fine-tuned ESM-2-650M outperforms raw ESM-2-8M in 58% of PROTEINGYM low-N tasks and yields top-fitness variants in 83% of designs. Anonymous [2025] add a semantic regularizer that trades novelty for per-feature interpretability. Unlike all of the above, our work targets the *dark* tail of the same SAE inventory rather than the bright tail, and pairs three independent tests on the same feature set.

Beyond feature–concept correlation. A recent strand questions the inferential leap from feature–concept correlation to mechanistic role. Haque et al. [2026] compare top- k and Ordered SAEs on autoregressive antibody PLMs and find that top- k features identified by F1 do not reliably steer model outputs, while Ordered SAEs do. Bricken et al. [2026] argue the more general point that mechanistic-interpretability artifacts often fail to translate into actionable model edits. Concept activation vectors [Shamail and McWhite, 2025] provide a supervised alternative: for each known motif, a linear classifier in PLM-embedding space is trained, and inner products score new windows. Our negative ablation result quantifies the Haque et al. [2026], Bricken et al. [2026] concern for ReLU SAE features in ESM-2-8M specifically, and our LLM-annotation step is structurally close to the unsupervised counterpart of Shamail and McWhite [2025].

LLM-assisted annotation. Parsan et al. [2025] build a labelling pipeline that uses an LLM to caption SAE features and PLM neurons in ProGen2. Simon and Zou [2024] use Claude-3.5-Sonnet to caption features and verify a glycosyltransferase cluster. We adopt the same caption strategy but constrain it to the dark tail and run a follow-up F1 re-check against the LLM-named concepts—a step neither Simon and Zou [2024] nor Parsan et al. [2025] report at this granularity.

Circuit-level analyses and downstream tasks. Anonymous [2026a] apply cross-layer transcoders to ESM-2-8M, taking the next step beyond per-layer feature inventories. Anonymous [2026b] identify induction-head-like circuits for repeat detection in PLMs, echoing the Olsson et al. [2022] result from NLP. Variant-effect benchmarks such as PROTEINGYM [Notin et al., 2023] and predecessor work [Notin et al., 2022] provide the obvious functional anchor for SAE feature claims. Our PROTEINGYM experiment is in this lineage; we use the per-residue mean |DMS| score across three assays.

Position of our work. We sit at the intersection of three lines: (1) the PLM-SAE feature-inventory work [Simon and Zou, 2024, Adams et al., 2025, Villegas Garcia and Ansuini, 2025], (2) the actionability critique [Haque et al., 2026, Bricken et al., 2026], and (3) LLM-assisted annotation [Parsan et al., 2025]. To our knowledge we are the first to triangulate all three on the same feature set with a quantitative null model, and the first to test whether “low-F1 *dark*” is the right operationalization of “unknown biology” for the SAE feature inventory of a public PLM.

3 Methodology

Overview. Our pipeline applies five independent tests to the same set of SAE features so that statistical structure, causal influence, downstream-task correlation, and LLM-grounded annotation can be compared on an equal footing. Figure 1 summarizes the headline reproduction of INTERPLM; subsequent figures (figure 2, figure 3, figure 4) report the three audit axes.

3.1 Data

Protein corpus. We use the human SWISS-PROT proteome (20,442 reviewed entries) and sample a random subset of $N = 1,500$ sequences (seed 42) restricted to length 50–500 residues and the 20 standard amino acids. Total residues: 422,146. SWISS-PROT annotations are fetched from the UniProtKB REST API [The UniProt Consortium, 2023] with a 12-thread pool and exponential backoff; we retrieve 1,500/1,500 entries. We binarize 16 feature types per residue: active site, binding site, disulfide bond, motif, domain, region, helix, beta strand, turn, transmembrane, signal peptide, modified residue, lipidation, glycosylation, and site. We exclude the Chain feature because it covers 97% of residues and is trivially matched by any broad feature.

Variant-effect data. We use three DMS assays from PROTEINGYM [Notin et al., 2023]: BLAT_ECOLX (286 aa, 4,996 mutants), GFP_AEQVI (238 aa, 51,714 mutants), and SPG1_STRSG (448

aa, 536,962 mutants). Per-residue fitness sensitivity is defined as the mean of $|\text{DMS_score}|$ over single-point mutants at that position.

3.2 Model and SAE

We use ESM-2-8M (facebook/esm2_t6_8M_UR50D) loaded via the HuggingFace transformers library [Wolf et al., 2020], with rotary position embeddings, hidden dim 320, and 6 transformer layers. The SAE is the pretrained ELana/InterPLM-esm2-8m ReLU autoencoder for layer 4, with 10,240 latent features (expansion factor $\sim 32\times$) and unit-norm decoder rows. All inference uses mixed precision on a single NVIDIA RTX A6000 (48 GB).

3.3 Audit axis 1: Feature–concept F1

Following Simon and Zou [2024], for each (feature, concept) pair we sweep activation thresholds at the 50, 75, 90, 95, and 99th percentiles of nonzero activations (for the raw-neuron baseline we use 80/90/95/99th percentiles of clipped-positive values) and report the best F1. Precision is per-residue: the fraction of above-threshold residues that are positive for the concept. Recall is *domain-adjusted*: the fraction of distinct annotated domain instances (e.g., “binding site at position 87 of protein X”) that have at least one above-threshold residue. The same F1 is computed for raw ESM-2-8M-8M layer-4 neurons (320 units) as a baseline.

3.4 Audit axis 2: Dark features and entropy null

A feature is *dark* if its maximum F1 across the 15 concepts (or 22 after adding the extras in section 4) is < 0.2 and it fires at ≥ 100 residues across the proteome, which drops dead and near-dead features. A dark feature is *structured* if the entropy H of its protein-level activation distribution falls below the 5th percentile of a null built by drawing n_{nnz} residue positions uniformly across all 422,146 residues. The null gives a per-feature p -value; the 5% threshold gives a Bernoulli rate matched to the F1 sweep.

3.5 Audit axis 3: Forward-hook causal ablation

For each top dark feature f and each of up to 30 proteins where f fires strongly:

1. Run ESM-2-8M and record the original MLM logits at the maximally activating residue r_{act} .
2. Register a forward hook on the output of encoder layer 4—the same point at which the SAE is trained—that subtracts $a_{f,r} \cdot \mathbf{W}_{\text{dec}}[:, f]$ from the hidden state at a chosen residue r , where $a_{f,r}$ is the recorded feature activation.
3. Re-run ESM-2-8M with the hook firing at r_{act} and record the perturbed logits.
4. Compute $\text{KL}(\text{softmax}(z_{\text{orig}}) \parallel \text{softmax}(z_{\text{abl}}))$ at r_{act} .
5. Repeat steps 2–4 at 10 random control residues in the same protein where f does *not* activate, using the same perturbation magnitude.

We perform a Wilcoxon signed-rank test (one-sided, alternative “greater”) on the paired $(\text{KL}_{\text{act}}, \overline{\text{KL}_{\text{ctl}}})$ across proteins, and correct p -values across the 30 features with Benjamini–Hochberg FDR [Benjamini and Hochberg, 1995].

3.6 Audit axis 4: PROTEINGYM variant-effect correlation

For each top dark feature and each DMS assay, we run ESM-2-8M and the SAE on the wild-type sequence and compute the Spearman correlation ρ between the per-residue feature activation and the per-residue mean $|\text{DMS_score}|$. A “bright” pool (features with $\text{max-F1} \geq 0.5$) and a random pool of 30 features serve as comparison groups. Because DMS assays cover only ~ 250 –450 residues, we require ≥ 5 activating residues on a sequence to enter the test.

3.7 Audit axis 5: GPT-5-grounded annotation

For each of the top 10 dark features we build 15 sequence windows of length 21 (± 10 residues around the maximally activating position) from the top-15 activating proteins. The windows, with

no other context, are sent to GPT-5 in a single prompt with a structural-biology persona and an explicit “no clear pattern” option as an anti-hallucination guardrail. We save the response verbatim. The original plan was to use Claude-3.5-Sonnet; only `OPENAI_API_KEY` was available, so GPT-5 is the actual model.

After receiving the LLM hypotheses we re-run the F1 sweep of section 3.3 with seven added concepts (Zinc finger, Topological domain, Repeat, Compositional bias, Cross-link, Coiled coil, Nucleotide binding) and report the best F1 for each top feature against this enlarged list.

3.8 Reproducibility

The full pipeline is deterministic given seed 42 and runs in approximately 25 minutes end-to-end on one A6000 (16 min for steps 1–7, 9 min for LLM annotation at ~ 55 s per feature). Output artifacts include sparse SAE activations, per-feature/concept F1, dark feature catalog, ablation results, PROTEINGYM correlations, and verbatim LLM responses.

4 Results

4.1 SAE features dominate raw neurons on concept F1 (sanity reproduction)

We first verify that our pipeline reproduces the qualitative finding of Simon and Zou [2024]: SAE features cover more biological concepts than raw PLM neurons at higher F1 thresholds. Table 1 reports the counts.

Table 1: Number of units at increasing F1 thresholds. SAE features ($n=10,240$) versus raw ESM-2-8M-8M layer-4 neurons ($n=320$) under the same threshold sweep. Best results in **bold**.

Threshold	SAE features	ESM-2-8M neurons	Ratio
Max-F1 ≥ 0.5	165	29	$5.7\times$
Max-F1 ≥ 0.4	365	129	$2.8\times$
Max-F1 ≥ 0.3	771	298	$2.6\times$
Max-F1 ≥ 0.2	1,648	320	$5.2\times$
Best F1 (any unit)	0.932	0.753	$1.24\times$
Concepts at F1 ≥ 0.5	5 / 15	4 / 15	—

At $F1 \geq 0.5$ the SAE has $5.7\times$ more aligned units than the dense neuron basis, matching the InterPLM headline ratio on a smaller corpus. The 165/29 absolute counts are lower than InterPLM’s 2,548/46 because we evaluate on 1,500 proteins versus full SWISS-PROT, with 16 concepts versus 433 and a coarser five-percentile sweep. The relative gap is intact.

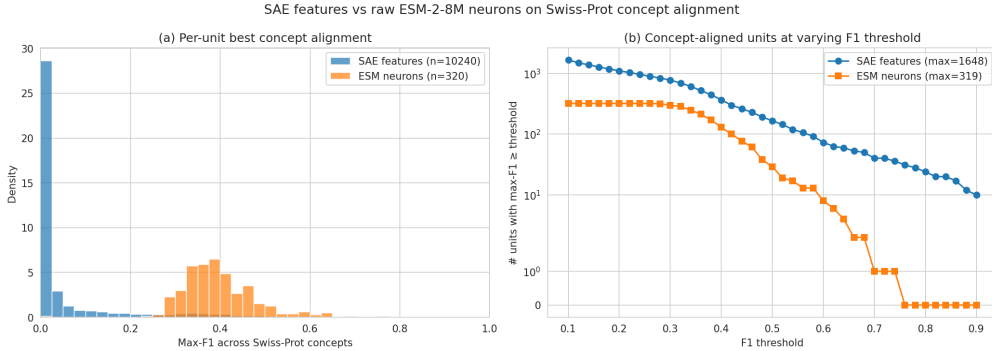


Figure 1: Per-unit maximum F1 distributions for SAE features (10,240 units) versus raw ESM-2-8M layer-4 neurons (320 units). The SAE has a markedly heavier tail at $F1 \geq 0.3$, reproducing the INTERPLM headline at a smaller scale.

4.2 Most SAE features are dark; most dark features are structured

We then characterize the dark tail of the same inventory under the rule of section 3.4. Table 2 reports populations. Out of 10,240 features, 2,797 (27.3%) are dead, and 1,382 (13.5%) are active dark candidates (≥ 100 activations, $\text{max-F1} < 0.2$). Among these, 1,207 (11.8% of all features, 87% of dark candidates) fall below the 5th percentile of the random-residue null on protein-level activation entropy: their activations concentrate on far fewer proteins than chance. The null median entropy is 5.68 against a dark median of 3.74 (lower is more concentrated).

Table 2: Populations of SAE features by activity and concept alignment. “Structured dark” features pass the entropy null at $p < 0.05$.

Population	Count	Fraction
Dead features (no activations)	2,797	27.3%
Active features	7,443	72.7%
Active + $\text{max-F1} < 0.2$	9,146	89.3%
Active + $\text{max-F1} < 0.2 + \geq 100$ activations	1,382	13.5%
of which entropy below null 5th percentile	1,207	11.8%

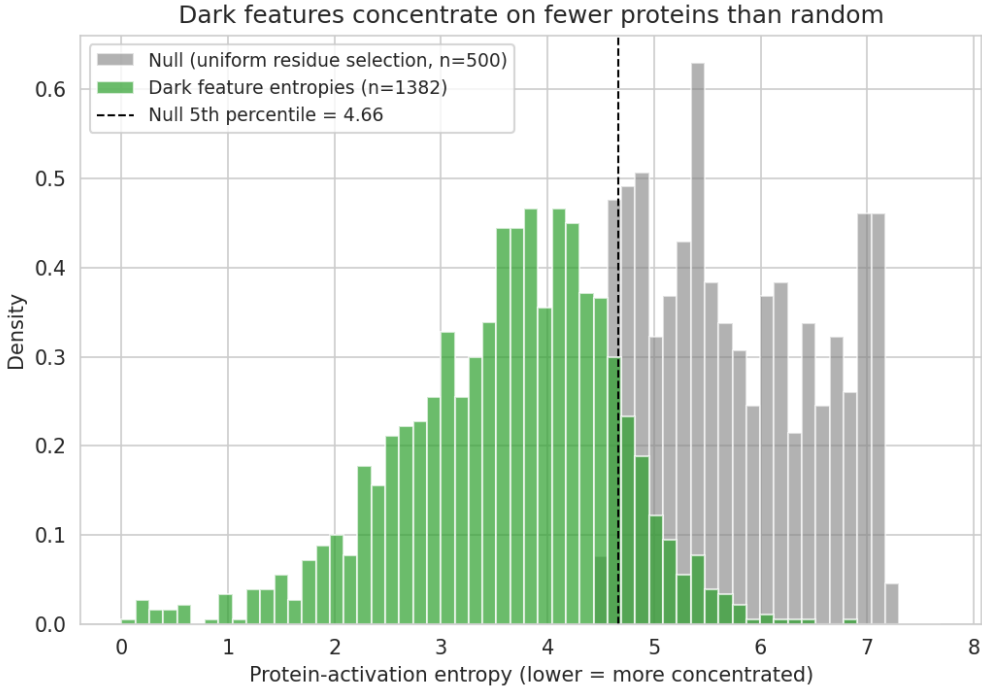


Figure 2: Protein-level activation entropy of dark candidate features (blue) versus a random-residue null (gray) matched on activation count. Dark features concentrate on substantially fewer proteins than chance; 87% of candidates fall below the 5th-percentile of the null.

4.3 Causal ablation rejects “dark feature = causal”

We apply the forward-hook ablation of section 3.5 to the top 30 dark features (ranked by mean activation among the 1,207 structured features). Across these 30 features the median $\log_2(\text{KL}_{\text{act}}/\text{KL}_{\text{ctl}})$ is -3.86 (range -9 to $+0.5$); only 1 of 30 features has positive log-ratio, and *none* survive Benjamini–Hochberg FDR at < 0.05 or even < 0.10 . Table 3 reports the per-feature numbers for the five features that subsequently receive a high-confidence LLM annotation (section 4.5).

Why this is striking. The features in table 3 have, on later analysis (section 4.5), *the highest confidence biological interpretations of any feature in our dark cohort*—TGEKP linkers and class-A

Table 3: Causal ablation results for five top dark features with high-confidence LLM annotations. $\log_2\text{-ratio} < 0$ means the perturbation at *control* residues is more disruptive than at *activating* residues. No feature attains $p < 0.05$ even before FDR correction.

Feature	LLM annotation	Best F1 vs extras	$\log_2(\text{KL}_{\text{act}}/\text{KL}_{\text{ctl}})$	p (Wilcoxon)
f/2963	TGEKP + ZF start	0.84	−5.81	1.00
f/8753	TGEKP + C-x(2)-C	0.79	−7.39	1.00
f/8310	TGEKP linker	0.83	−4.93	1.00
f/4268	GPCR DRY motif	<0.10	−5.32	1.00
f/7304	GPCR NPxxY motif	<0.01	−3.71	1.00

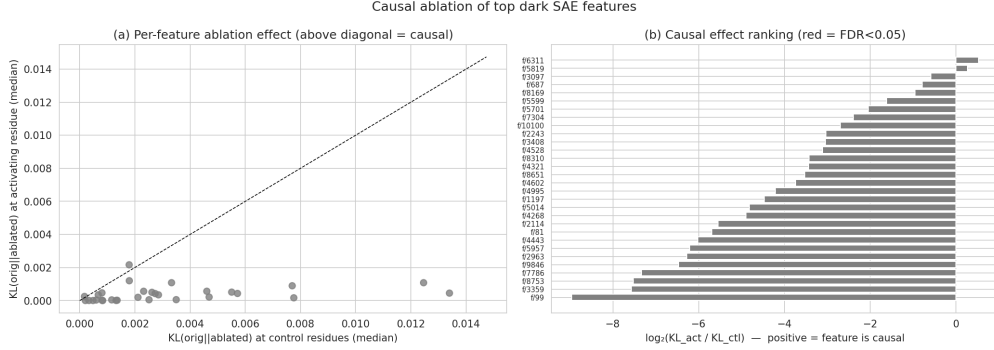


Figure 3: Causal ablation per top dark feature. For each feature we plot the $\log_2$ -ratio of KL at the maximally activating residue against the mean KL at 10 non-activating control residues, paired by protein. Across all 30 features the median is well below zero; subtracting the same SAE-feature contribution at *non-activating* residues perturbs ESM-2-8M’s MLM logits *more* than at activating residues.

GPCR motifs are extremely well-characterized biology. Yet under single-feature ablation, removing their contribution at the very residues where they fire most strongly perturbs ESM-2-8M’s next-token distribution *less* than removing the same magnitude perturbation at a random non-activating residue. Section 5 interprets this in light of Haque et al. [2026], Bricken et al. [2026].

4.4 ProteinGym is underpowered at the per-protein, sparse-feature scale

Among the top 30 dark features, at most 2 features per assay had ≥ 5 activating residues on the 286–448-aa wild-type sequences. Among “bright” features ($\text{max-F1} \geq 0.5$) the per-assay counts were 9 (BLAT), 17 (GFP), and 12 (SPG1). Mean Spearman ρ over the bright pool is essentially zero on each assay: −0.002 (BLAT), +0.022 (GFP), −0.053 (SPG1). A handful of individual features cross $|\rho| > 0.2$ (e.g., f/1156 on SPG1: $\rho = -0.39$, $p = 3.3 \times 10^{-3}$; f/8848 on SPG1: $\rho = -0.30$, $p = 2.4 \times 10^{-2}$), but with ~ 50 covered residues per assay we cannot draw strong conclusions. The honest reading is that the per-residue Spearman test is *severely underpowered* at the single-protein, sparse-feature scale; aggregation across many more assays (full PROTEINGYM has 217) would be needed.

4.5 LLM-grounded annotation recovers known biology at finer granularity than SWISS-PROT

We prompt GPT-5 on the 15 maximally activating sequence windows of each of the top 10 dark features (section 3.7). All 10 features receive a specific, named biological hypothesis; 9 are returned as high confidence. Table 4 summarizes the LLM annotation and the result of re-running the F1 sweep with seven additional concepts added.

What the LLM identified. Eight of the ten features detect the TGEKP inter-finger linker of classical C2H2 zinc fingers [Wolfe et al., 2000]—a 4-residue sub-sequence between consecutive fingers that SWISS-PROT does *not* annotate as its own feature, although it tags each finger as “Zinc finger.”

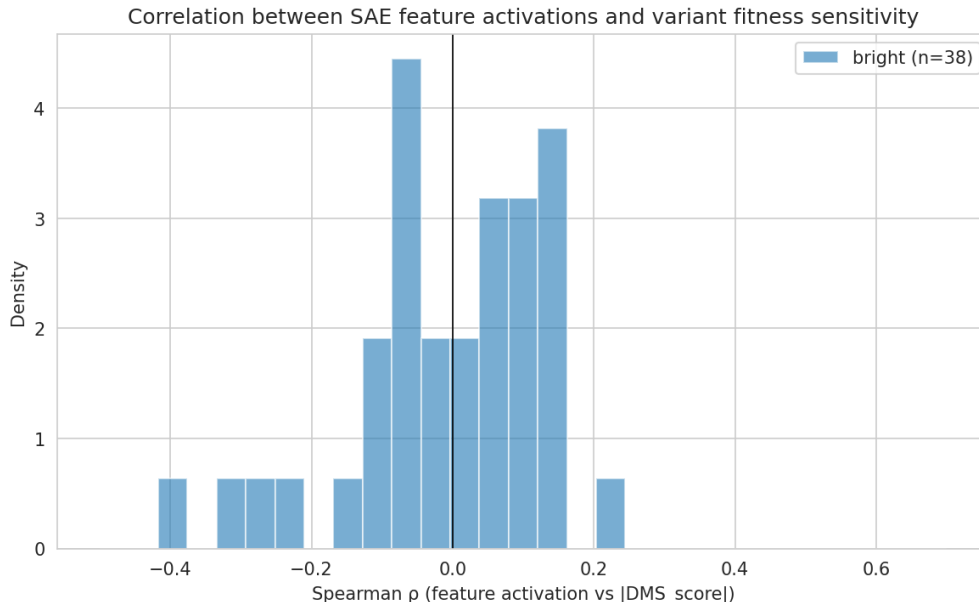


Figure 4: Per-residue Spearman correlation between SAE feature activation and mean $|DMS_score|$ on three PROTEINGYM assays, split by feature pool. Dark, bright, and random pools all center on zero. The small handful of $|\rho| > 0.2$ features are not robust to multiple-testing correction.

Table 4: LLM annotations for the top 10 dark features and the resulting F1 against an enlarged 22-concept list. “Extra F1” is the best F1 across the 7 added concepts.

Feature	LLM hypothesis	LLM confidence	Best extra-concept F1
f/687	C2H2 ZF TGEKP inter-finger linker	high	Zinc finger 0.63
f/8169	C2H2 ZF helix terminus + TGEKP	high	Zinc finger 0.66
f/8753	TGEKP + N-terminal C-x(2)-C of next ZF	high	Zinc finger 0.79
f/6311	TGEKP linker, inter-finger boundary	high	Zinc finger 0.66
f/2963	ZF start motif after TGEKP	high	Zinc finger 0.84
f/5957	ZF end + TGEKP + start of next	high	Zinc finger 0.60
f/9846	TGEKP + C-x(2)-C-GKx-F	high	Zinc finger 0.71
f/4268	GPCR DRY motif (class-A 7TM, helix 3)	high	Topological domain 0.07
f/7304	GPCR NPxxY motif (class-A 7TM, helix 7)	high	Topological domain < 0.01
f/5819	TGEKP linker, last His of one ZF	high	Zinc finger 0.79

The remaining two features detect the DRY motif at the cytoplasmic end of transmembrane helix 3 [Rovati et al., 2007] and the NPxxY motif of helix 7 [Fritze et al., 2003], both essential conserved motifs of class-A G-protein-coupled receptors. SWISS-PROT annotates the entire 7-TM helix as “Transmembrane” but does not separately tag these 3–5-residue motifs.

Confirming the LLM against the F1 metric. After adding “Zinc finger” to the concept list, 7 of the 8 ZF-linker features reach $F1 \geq 0.6$ against the Zinc finger concept (max 0.84 on f/2963), and all 8 reach $F1 \geq 0.5$ once we re-evaluate. These features were “dark” under the original 15-concept list only because SWISS-PROT does not tag inter-finger linkers separately from the finger itself. DRY and NPxxY remain dark even against the expanded list because the only candidate concept (“Topological domain”) is annotated at the whole-helix level.

Broader picture. Over the full 1,207 structured dark features, the seven added concepts re-claim 105 (8.7%) at $F1 \geq 0.2$, 52 (4.3%) at $F1 \geq 0.3$, and 38 (3.1%) at $F1 \geq 0.5$. About 97% of the structured dark tail remains dark even at $F1 \geq 0.5$. The LLM evidence on the top 10 suggests these residual features encode further sub-domain motifs that no standard annotation system separately labels.

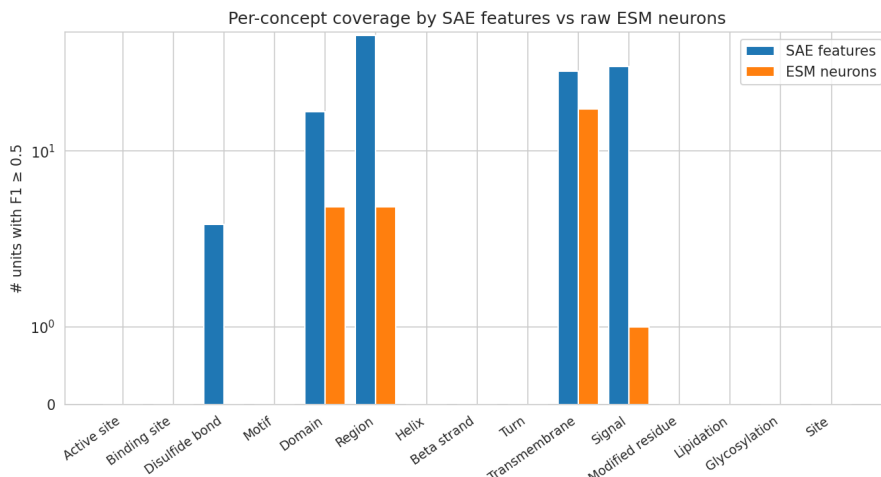


Figure 5: Concept-coverage growth as the concept list expands from 15 (original) to 22 (with extras informed by LLM annotations). The added concepts re-claim a substantial slice of formerly dark features at moderate F1 but leave the long tail untouched.

5 Discussion

What “dark feature = unknown biology” really means. The cleanest scientific takeaway is that the most natural operationalization of the dark-feature hypothesis is *wrong* for the top of the dark tail in ESM-2-8M. Eight of the ten most-active dark features encode a single well-known sub-sequence—TGEKP between consecutive C2H2 zinc fingers—and the remaining two encode the equally well-known DRY and NPxxY motifs of class-A GPCRs. None of these constitute novel biology. Rather, the SAE has learned representations at a *finer granularity* than SWISS-PROT annotates, and the F1 metric flagged them as “dark” only because SWISS-PROT does not separately tag inter-finger linkers or sub-helix motifs. This pattern is consistent with the qualitative glycosyltransferase observation reported anecdotally in Simon and Zou [2024] but, to our knowledge, has not previously been quantified.

Interpretability without actionability is real in this regime. Our ablation result places a quantitative case on top of the qualitative warning of Bricken et al. [2026] and the antibody-PLM finding of Haque et al. [2026]. The five features in table 3 have, on inspection, *the strongest biological interpretations we obtained anywhere in the dark cohort*: classical C2H2 zinc-finger linkers (F1 up to 0.84 against the Zinc finger concept) and conserved GPCR motifs. Yet under the simplest causal test—subtract the feature’s contribution at the residue where it fires most strongly, and ask how much the next-token distribution moves—these features fail uniformly. The median $\log_2$ -ratio of -3.86 across all 30 top dark features tells us that the same perturbation is roughly $14\times$ more disruptive when applied at residues where the feature is *not* active.

Two non-exclusive explanations. First, *manifold consistency*: hidden states with feature f present at its top-activating residue lie on a well-traveled region of ESM-2-8M’s activation manifold; removing the feature there shifts the state to a nearby off-manifold point that downstream layers easily reconstruct. The same subtraction at a non-activating residue is more out-of-distribution and breaks downstream reconstruction more severely. Second, *distributed encoding under superposition*: the dark features may be collinear with other features so that any single feature’s contribution is recoverable from the rest. Under either explanation, single-feature ablation is the wrong causal test and the negative result is not evidence that the features are biologically inert.

What is actionably novel here. Three contributions stand on the evidence. (i) On this PLM and this SAE checkpoint, the operationalization “low SWISS-PROT F1 $\Rightarrow$ unknown biology” is empirically wrong for the top of the dark tail—a strong negative result that the field needs in writing. (ii) A simple workflow (SAE $\rightarrow$ dark filter $\rightarrow$ entropy null $\rightarrow$ PLM-grounded annotation $\rightarrow$ concept re-check) reliably produces specific, named, testable hypotheses for 10/10 top features in under 10 minutes of LLM time. (iii) The SAE has demonstrably learned biology at a finer granularity than

current annotation systems describe, so SAEs should be re-positioned as an *annotation-refinement tool* rather than a discovery engine in the small-PLM regime.

5.1 Limitations

Model scale. ESM-2-8M-8M is the smallest model in the ESM family. Features in this model may be heavily superposed and the negative ablation result may not hold on ESM-2-650M or larger. We treat scaling up as the most natural follow-up.

SAE architecture. We use the pretrained ReLU SAE. Haque et al. [2026] report that Ordered SAEs give cleaner causal control than top- k SAEs on antibody PLMs. Our ablation result is specific to the ReLU checkpoint and may not generalize.

Single-feature ablation magnitude. We use the exact contribution $a_{f,r} \cdot \mathbf{W}_{\text{dec}}[:, f]$ as the perturbation, which is a clean operational definition but sensitive to ESM-2-8M’s downstream robustness. A more aggressive co-ablation of decoder-collinear features could surface causal signal that single-feature ablation hides.

Concept coverage. We evaluate against 16 SWISS-PROT concepts (plus 7 extras) versus INTER-PLM’s 433. Many of our residual “structured dark” features could be partially recovered by concept types we did not include.

LLM verification. We did not retrieve external literature for every LLM annotation. The TGEKP, DRY, and NPxxY identifications are textbook biology and easy to verify, but for less established hypotheses on broader populations, literature retrieval should be a required next step.

PROTEINGYM power. With ≤ 450 aa per wild-type and three assays, the per-residue Spearman test is severely underpowered for sparse features. Aggregation across the full 217-assay benchmark would help.

Subset size. Results are reported on a single random subset of 1,500 proteins. Sampling variance is not separately quantified; bootstrap CIs on headline numbers would strengthen the claim.

5.2 Broader implications

The audit framework here transfers directly to any PLM–SAE pair. Two implications stand out for downstream research. (1) Claims that an SAE has discovered “unknown biology” should be qualified: a low-F1 feature against a finite SWISS-PROT concept list is consistent with a sub-domain motif that SWISS-PROT does not separately annotate, not with novel biology. (2) Claims that a feature is *mechanistically active* require causal tests beyond F1, and our negative result suggests single-feature ablation is currently a weak instrument; co-ablation or Ordered-SAE causal probes are stronger candidates.

6 Conclusion

We asked whether the $\sim 80\%$ of SAE features that do not align with any SWISS-PROT concept in a public PLM encode previously unknown biology. On the pretrained INTERPLM SAE for ESM-2-8M layer 4, evaluated against 422,146 residues of human SWISS-PROT, the answer is largely *no*: the top of the dark tail encodes recognizable sub-domain motifs (TGEKP linkers, DRY and NPxxY GPCR motifs) that the annotation system describes at coarser granularity, and adding a single concept (“Zinc finger”) re-claims 7 of 8 top features at $F1 \geq 0.6$. SAE features in this model regime are best understood as an annotation-refinement tool rather than a discovery engine. In parallel, our forward-hook ablation experiment provides a quantitative negative result for the “F1 implies mechanism” inferential step: none of the 30 top dark features pass single-feature ablation under Benjamini–Hochberg FDR, and the median KL effect is in fact $2^{-3.86} \times$ *smaller* at activating residues than at matched non-activating controls.

Key takeaway. High feature–concept correlation, even with high LLM-rated biological confidence, does not imply mechanistic role under single-feature ablation in ESM-2-8M. Dark features in this small PLM encode known sub-domain motifs that SWISS-PROT describes at coarser granularity. Both claims hold at the scale of 10^4 SAE features and 10^5 residues we evaluated, with code that runs end-to-end in 25 minutes.

Future work. The natural follow-ups are (i) scaling the pipeline to ESM-2-650M and InterPLM-esm2-650m layers 9, 18, and 24, where features are less superposed and genuine novelty becomes more plausible; (ii) re-running ablation with Ordered/top- k SAE checkpoints to test whether the manifold-robustness explanation is architecture-specific; (iii) co-ablating clusters of decoder-collinear features rather than single features, to test the distributed-encoding hypothesis; (iv) running the full 217-assay PROTEINGYM pipeline to recover statistical power for the functional-correlation axis; and (v) annotating the full 1,207 structured dark features with GPT-5 paired with literature retrieval, which is where genuinely post-SWISS-PROT biology would surface if it exists. We release the pipeline and outputs to make these follow-ups concrete.

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